

3 Hydrogen bonding has many important roles in organisms.

- (a) State **and** explain the importance of hydrogen bonding for the specific heat capacity of water **and** for the latent heat of vaporisation of water.

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..... [3]

- (b) Describe **and** explain the role of hydrogen bonding in the movement of water in xylem vessels in plants.

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..... [3]

(c) Figure 3.1 shows part of a DNA molecule.

The four bases in Figure 3.1 are labelled **W**, **X**, **Y** and **Z**.

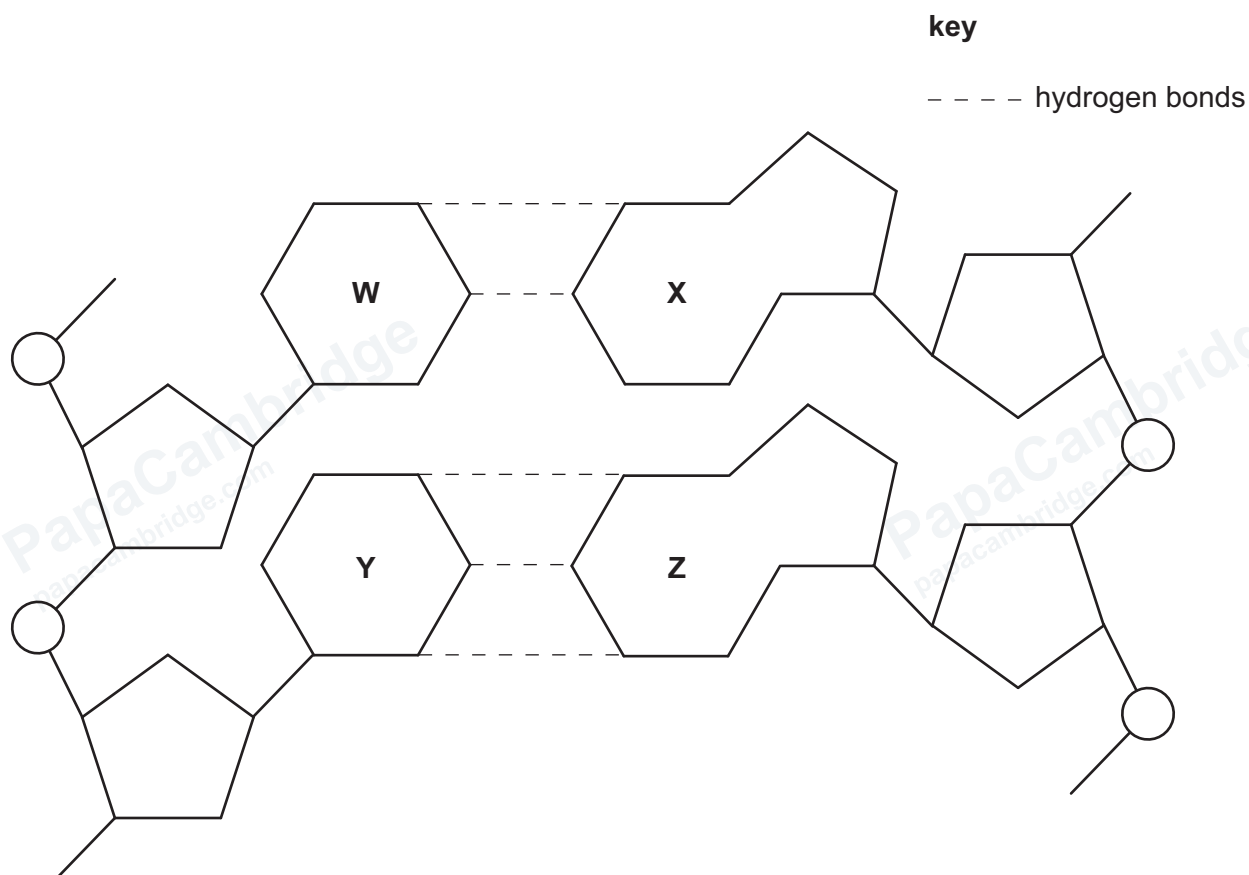


Figure 3.1

State the **names** of bases **W**, **X**, **Y** and **Z** in Figure 3.1.

W

X

Y

Z

[3]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(d) Table 3.1 lists three types of interaction that maintain the shape of protein molecules.

Complete Table 3.1 by placing a tick (✓) in the box if the type of interaction can affect the primary structure, secondary structure or tertiary structure of a protein.

Table 3.1

interaction	primary structure	secondary structure	tertiary structure
hydrogen bonds			
hydrophobic interactions			
ionic bonds			

[3]

[Total: 12]



Turn over